

Technical Descriptions and Process Analyses

Technical Descriptions and Process Analyses



COMMUNICATION AT WORK

The information technology scenario shows the importance of technical descriptions and process analyses in the engineering profession.

Alpha Beta Consulting (A/B/C) specializes in creating software for public service engineering purposes, such as storm water, water distribution, transportation and traffic control, parks and recreation, facility and building management, and inventory of fleets and equipment.

The city of Maple Valley, Texas, is interested in purchasing one of A/B/C's software suites related to storm water maintenance. The suite focuses on accurate and easy-to-use inventory and inspection tools for maintaining and improving storm water systems. A/B/C's software will help Maple Valley inspect and test all components of the city's storm water system including conduits, structures, detention basins, pump stations, and pumps.

Before A/B/C can sell the software to Maple Valley, the company's information technology director, Neeha Patel, must do the following:

- Determine the city's unique needs—Neeha will interview city residents, city council members, and

Learning Objectives

When you complete this chapter, you will be able to

- Understand that technical descriptions are often used in user manuals, product specifications, construction designs, reports, and sales literature
- Understand that process analyses tell how a tool, mechanism, piece of equipment, product, or service works
- Use the introduction of a technical description or process analysis to specify and define the topic; explain the topic's functions, capabilities, or processes; and include a list of components, parts, or equipment
- Know that graphics with callouts make a technical description and process analysis easier to understand for the reader
- Follow a spatial pattern of organization in a technical description
- Follow a chronological order of organization when describing how a process works
- Use words for descriptions and processes that are photographic and/or impressionistic
- Use headings and subheadings to organize the content of your description or process analysis
- Use line drawings, schematics, CAD/CAM drawings, photographs, architectural renderings, clip art, exploded views, or sectional cutaway views of your topic for the technical description or process analysis
- Evaluate your description or process by using the checklist

city government officials. To accomplish this goal, Neeha will create a questionnaire focusing on infrastructure needs. Then, through a combination of hard-copy mailers and online social media (Maple Valley's Facebook site and Twitter account), Neeha will collect constituent responses.

- Research software options—Based on the interviews, Neeha will determine if the city's current software is failing to meet the city's needs and what A/B/C software options exist.
- Create or adapt software to meet the city's technical specifications—If new software is needed, Neeha will work with her colleagues in IT to customize software for Maple Valley.
- Write technical descriptions of the hardware—Based on her analysis of hardware requirements, Neeha will write the technical descriptions of hardware needed to run the software. These descriptions will accompany the operations manual A/B/C will give Maple Valley employees.
- Provide process analyses—To clarify how the software will solve Maple Valley's infrastructure problems, Neeha will write a process analysis. This process analysis will show how the software will improve storm water, transportation, and traffic flow.

Once Neeha creates this documentation, she can provide it to Maple Valley's Planning Commission, which will decide whether or not to purchase the software and work with A/B/C.

Defining Technical Descriptions

A *technical description* is a part-by-part depiction of the components of a mechanism, tool, or piece of equipment. Technical descriptions are important features in several types of technical communication.

Types of Technical Descriptions

Technical writers use technical descriptions in a variety of documents.

Operations Manuals

Manufacturers often include an *operations manual* in the packaging of a mechanism, tool, or piece of equipment. This manual helps the end-user construct, install, operate, and service the equipment. Operations manuals also can contain technical descriptions.

Technical descriptions provide the end-user with information about the mechanism's features or capabilities. For example, this information may tell the user which components are enclosed in the shipping package, clarify the quality of these components, specify what function these components serve in the mechanism, or allow the user to reorder any missing or flawed components. Following is a brief technical description found in an operations manual.

EXAMPLE ►

The Modern Electronics Tone Test Tracer, Model 77A, is housed in a yellow, high-impact plastic case that measures 1¼ inch { 2 inch { 2¼ inch, weighs 4 ounces, and is powered by a 1604 battery. Red and black test leads are provided. The 77A has a standard four-conductor cord, a three-position toggle switch, and an LED for line polarity testing. A tone selector switch located inside the test set provides either solid tone or dual alternating tone. The Tracer is compatible with the EXX, SetUp, and Crossbow models.

Product Demand Specifications

Sometimes a company needs a piece of equipment that does not currently exist. To acquire this equipment, the company writes a *product demand* specifying its exact needs. The following product demand specification example is written for a high-tech audience. It assumes knowledge on the part of the reader, such as definitions for “high-speed” and “deep-hole,” which a lay reader would not understand. In addition, the high-tech abbreviation “AISI” is not defined.

EXAMPLE ►

Subject: Pricing for EDM Microdrills

Please provide us with pricing information for the construction of 50 EDM Microdrills capable of meeting the following specifications:

- Designed for high-speed, deep-hole drilling
- Capable of drilling to depths of 100 times the diameter using 0.012-inch to 0.030-inch diameter electrodes
- Able to produce a hole through a 1.000-inch-thick piece of AISI D2 or A6 tool steel in 1.5 minutes, using a 0.020-inch diameter electrode

We need your response by January 13, 2014.

Study Reports Provided by Consulting Firms

Companies hire a consulting engineering firm to study a problem and provide a descriptive analysis. The resulting *study report* is used as the basis for a product demand specification requesting a solution to the problem. One firm, when asked to study crumbling cement walkways, provided the following technical description in its study report.

The slab construction consists of a wearing slab over a ½-inch-thick waterproofing membrane. The wearing slab ranges in thickness from 3½ inches to 8½ inches, and several sections have been patched and replaced repeatedly in the past. The structural slab varies in thickness from 5½ inches to 9 inches with as little as 2 inches over the top of the steel beams. The removable slab section, which has been replaced since original construction, is badly deteriorated and should be replaced. Refer to Appendix A, Photo 9, and Appendix C for shoring installed to support the framing prior to replacement.

◀ EXAMPLE

Construction Design

Prior to building any structure, architectural companies must clarify the construction design for the city and the clients. This requires a legal technical description of the property limits (location and dimensions). Once this description is written and drawn, the architectural company will use the text and graphics for a variety of purposes. These can include presentations of the design to a city council or homeowner's association, requests for a change of zoning, requests for special use permits, and re-platting of the site if a property line must be relocated. Finally, based on the legal technical description, the architectural firm can obtain city approval. See Figure 1 for a layout and landscaping plan and the legal technical description that accompanies it.

Sales Literature

Companies want to make money. One way to market equipment or services is to describe the product. Such descriptions are common in sales letters, proposals, and on Web sites. Figure 2 is a technical description in sales literature from Hewlett-Packard.

Defining Process Analysis

What is a process analysis? A process analysis is comparable to an instruction. Instructions provide a step-by-step explanation of how to do something. For example, in an instruction, you explain how to change the oil in a car, how to connect a printer to a computer, how to make polenta, or how to put together a child's toy. In an instruction, the audience wants to know how to do a job. A process analysis, in contrast, focuses not on how to do something but on how something works. For instance, a process analysis might explain how viruses attack our bodies, how airbags save lives, how metal detectors work, or how e-mail messages are transmitted.

Examples of Process Analyses

A discussion of process is a common part of many technical descriptions.

- **Engineering.** Dubai, second largest emirate of the United Arab Emirates, plans to develop several islands off its coast. To increase tourism, these islands will have

DOT.COM UPDATES

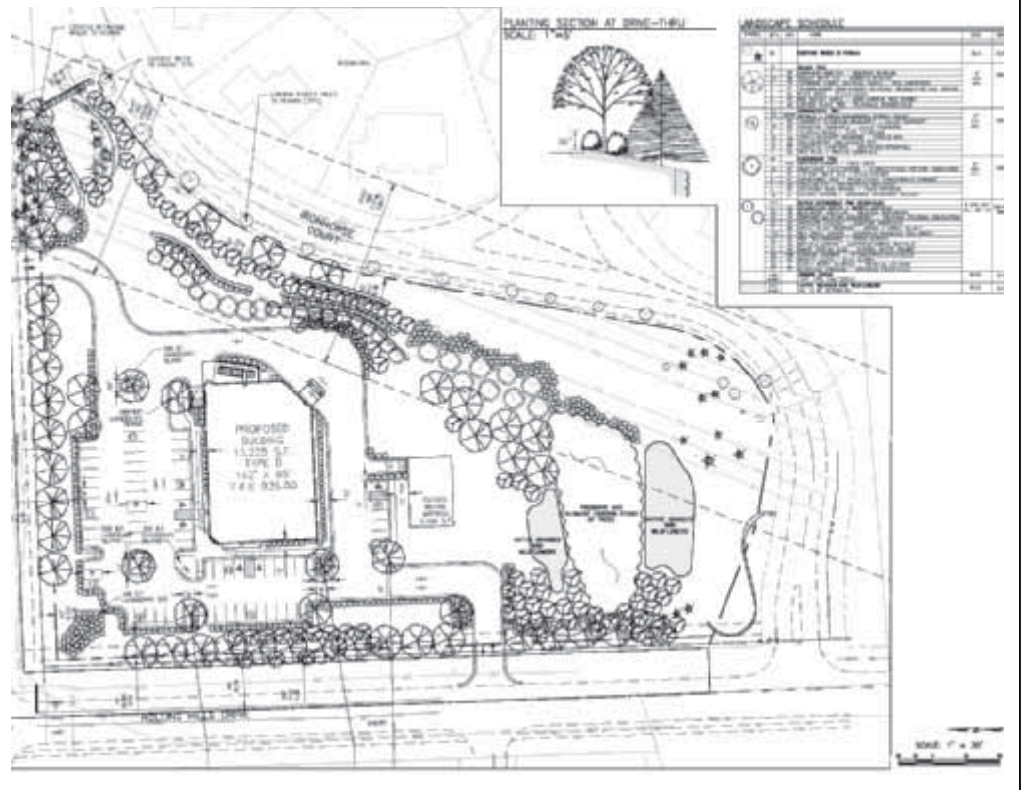
For examples of technical descriptions, check out the following links:

- **Palm Pre Smartphone technical specifications**
<http://www.palm.com/us/products/phones/pre/index.html>
- **iPod Nano technical specifications**
<http://www.apple.com/ipodnano/specs.html>
- **Technical specifications for Garmin's GPS system**
http://www.garmin.com/manuals/425_TechnicalSpecification.pdf
- **Technical specifications for the Boeing 757**
<http://www.boeing.com/commercial/757family/technical.html>
- **Original Xbox technical specifications**
<http://www.xbox.com/en-US/support/familysettings/resources/consolespecs.htm>
- **Nintendo Wii technical specifications**
<http://www.nwiizone.com/nintendo-wii/nwii/nintendo-wii-technical-specifications/>

FIGURE 1 Layout and Landscaping Plan with Legal Technical Description for a High-Tech Audience

This landscape plan describes the construction design for section 10, township 14, range 25, located in the city of Hartford, Johnson County, Washington. The plot of land includes the following coordinates and distances:

- north $87^{\circ}38'42''$ east, along the north line of the northwest quarter, a distance of 539.17 feet
- south $2^{\circ}21'18''$ east, along a line perpendicular to the north line of the northwest quarter, a distance of 60.00 feet
- west $51^{\circ}29'37''$, a distance of 314.56 feet, to a point of tangency
- south $09^{\circ}56'20''$ west, along the westerly right-of-way line, a distance of 264.15 feet, to a point of curvature
- east $59^{\circ}29'39''$, a distance of 129.80 feet, to the most easterly corner of tract "c" and east of the right-of-way line of Rolling Hills Drive



Source: Courtesy of Landplan

both private residences and hotels. Your engineering company has been hired to build bridges connecting the islands to the coastline. Your options include beam bridges, arch bridges, and suspension bridges. To clarify which type of bridge will best meet your client's needs, you provide a PowerPoint presentation explaining how each bridge works.

FIGURE 2 Technical Description for a Lay Audience

Source: Courtesy of Hewlett-Packard

The title identifies the equipment to be described.

Impressionistic words such as “quick access,” “easy troubleshooting,” and “User-friendly control panel” are used for sales purposes to the lay audience.

Numbered callouts refer to the text and help the readers locate parts of the equipment.

- **Automotive sales.** A potential customer wants to buy a new car. This customer is concerned about the environment and is interested in buying a hybrid, knowing that these cars are energy efficient and ecologically friendly. However, the customer doesn’t understand how hybrids work. Your job is to explain the process of how gasoline-electric hybrid cars work.
- **Biomedical technology.** You and a team of scientists have been working on new treatments for diabetes, a health problem in the United States that affects approximately 20.8 million Americans. One possible treatment is pancreatic islet transplantation. Islets are injected through a catheter into a diabetic’s liver. In time, the islets begin releasing insulin. This treatment has great potential for diabetics, but research is ongoing. You are writing a proposal requesting additional funding for research. In this proposal, you provide a process analysis of how diabetes affects a patient and how the pancreatic islet transplantation will work to treat this disease.

SPOTLIGHT

How to Use Technical Descriptions in Landscape Architecture

Landplan, a multidisciplinary firm that specializes in surveying, civil engineering, landscape architecture, and planning, designs sites that conform to local city and county codes. As a landscape architect with Landplan, **Andrew Buchwitz** works with one of their larger corporate accounts to develop commercial sites. According to Andrew, “I lay out the best possible location for the building, parking, and landscaping while conforming to the local municipalities’ City Development Code.”

Communicating effectively is a major part of Andrew’s job. He writes technical descriptions that accompany his architectural renderings, and he makes oral presentations at city planning commission meetings, city council meetings, and neighborhood meetings.

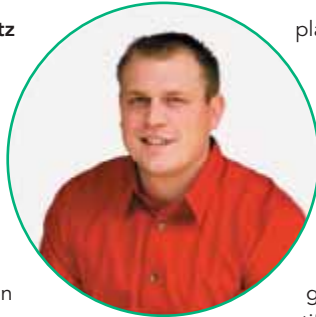
Andrew says that Landplan has “to write legal descriptions depicting the property size and location.” Technical and legal descriptions are important for every project Andrew designs to

- clarify the plan for the city and the client
- meet legal description/property limits
- meet city requirements for landscaping
- obtain city approval so the client can build

Andrew’s challenge is conforming to the local cities’ development codes while still meeting his clients’ needs. “Many times it becomes a compromise when the client wants the project to look a certain way, but the city does not,” states Andrew.

For example, the city planners could review the renderings, technical specifications, technical descriptions, and documents Andrew designs but refuse to accept the plans as they have been drafted. The city might suggest site revisions including additional landscape screening around trash enclosures and additional upgrades to the exterior such as brick rather than stucco. In those

instances, Andrew has to revise the plans to meet both the city and the client’s needs.



Every site is unique in that it requires a different plan of attack. This could include requesting a change of zoning, requesting a special use permit, or having to re-plat the site because a property line needs to be relocated. “Some of these require public notice. In this case, Landplan would have to send a mail notification to residents living in a 200- to 500-foot buffer area surrounding the site.”

According to Andrew, “Mail notices also usually require a neighborhood meeting for the general public. This is basically a Q & A session notifying the public of the changes that are being proposed for the particular area and hearing about any concerns they may have.”

At these neighborhood meetings, Andrew often makes oral presentations. According to Andrew, “Some of the challenges may be getting the general public to back our plan. If the city commission sees that the general consensus of the plans for the buffered public area is against this development, it may be even harder to get approval. The city may not like the proposed development and want us to have another meeting with city planners to come up with another layout or design.”

Once the City Planning Commission approves all of Andrew’s site revisions, the Commission takes the plans to the City Council Community Development Committee for final approval. Andrew then returns the approved documents to the client who gives the “go-ahead” for the project. Throughout the multi-step process, Andrew faces many challenges communicating both orally and in writing with the client, the city, and other concerned parties. Carefully written technical and legal descriptions, as well as effective oral presentations, help to ensure approval of Landplan’s proposed site development.

Figure 3 is a process analysis of why, when, and how automobile air bags inflate.

Criteria for Writing Technical Descriptions and Process Analyses

As with any type of technical communication, there are certain criteria for writing technical descriptions and process analyses.

Title

Preface your text with a title precisely stating the topic. This could be the name of the mechanism, tool, piece of equipment, landform, product, or service you are describing or analyzing.

FIGURE 3 Why, When, and How Does an Air Bag Inflate?

Air bags save lives. Driver and passenger air bags are designed to inflate in frontal or side crashes. Steering wheel, right front instrument panel, or side-panel air bags will not inflate on all occasions. If a car drives over a bump or if a crash is “minor,” such as hitting an object while driving at a slow speed, an air bag will not deflate. However, when cars hit walls (or trucks or cars or trees), air bags inflate to minimize injury and to save people’s lives.

Before Inflation

How does the air bag detect whether the car has hit a bump versus being involved in a collision?

The airbag system’s crash sensor can differentiate between head-on collisions and simple bumps in the road as follows:

1. A steel ball slides inside a smooth bored cylinder.
2. The ball is secured by a magnet or by a stiff spring. This inhibits the ball’s motion when the car encounters minor motion changes, such as bumps or potholes.
3. If the car comes to a dramatic and rapid stop, a force equal to running into a brick wall at about 10 to 15 miles per hour, such as in a head-on crash, the ball quickly moves forward. This closes a contact and completes an electrical circuit, which initiates the process of inflating the airbag.

Parts of an Air Bag

Air bag systems consist of the following:

- **Air bag:** made of thin nylon. A nontoxic powder (cornstarch or talcum powder) inside the air bag keeps it flexible and ensures the parts of the air bag do not stick together.
- **Air bag holding compartment:** the steering wheel, dashboard, seat, or door.
- **Sensor:** a device that tells the bag to inflate.
- **Inflator canister:** consisting of sodium azide (NaN_3), potassium nitrate (KNO_3), and silicon dioxide (SiO_2), which produce nitrogen gas (N_2).

During Inflation

In a severe impact, the air bag sensing system will deploy in milliseconds. The following occurs:

1. The air bag’s crash sensor triggers a switch that energizes a wire, sending electricity into a heating element in the propellant that releases gas from the inflator canister.
2. A pellet of sodium azide (NaN_3) is ignited, generating nitrogen gas (N_2).
3. A nylon air bag, folded into the dashboard, steering wheel, and/or side panels of the door, inflates at a speed of 150–250 mph, taking only about 40 milliseconds (about 1/20 of a second) for the inflation to be complete.

The introduction explains what the topic is, why it is important, and where the mechanism is located.

This technical description discusses the parts of the mechanism, the materials used, the location of these components, and the chemical compounds required for activation.

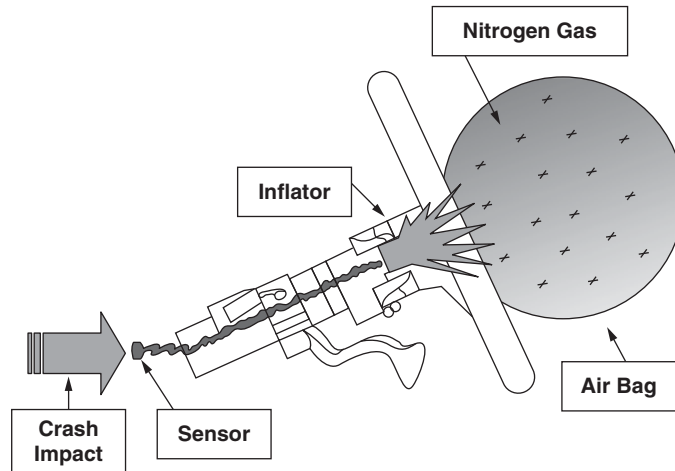
Here, providing the mechanism’s process, the text explains how an air bag works. Note the specificity of detail: 150–250 mph and 1/20 of a second.

(Continued)

FIGURE 3 Why, When, and How Does an Air Bag Inflate? (Continued)

4. The sodium azide inside the air bag produces sodium metal, which is highly reactive, potentially explosive, and harmful when in contact with eyes, nose, or mouth. To render this harmless, the sodium azide reacts with potassium nitrate (KNO_3) and silicon dioxide (SiO_2)—also inside the air bag—to produce silicate glass, a harmless and stable compound.

This illustration graphically depicts the process in action. When the crash sensor is triggered, it activates the inflator. Nitrogen gas explodes, inflating the air bag.



After Inflation

Air bag vents, minute holes in the bag, allow the deployed air bags to deflate immediately after impact. This ensures that the car's inhabitants do not smother.

Though air bags were first used in 1973, they have only been mandatory in all cars since 1998. Have air bags made a difference? Absolutely! The National Highway Traffic Safety Administration says air bags saved an estimated 1,043 lives in 1998 alone.

The conclusion sums up the process analysis by quantifying the significance of air bags as a means of saving lives.

Overall Organization

In the **introduction**, specify and define your topic, and explain the topic's functions, capabilities, or processes. You can also include a list of components, parts, or equipment.

EXAMPLE ►

Technical Description

The Apex Latch (#12004), a mechanism used to secure core sample containers, is composed of three parts: the hinge, the swing arm, and the fastener.

EXAMPLE ►

Technical Description and Process Analysis

The DX 56 DME (Distance Measuring Equipment) is a vital piece of aeronautical equipment. Designed for use at altitudes up to 30,000 feet, the DX 56 electronically converts elapsed time to distance by measuring the length of time between your transmission and the reply signal. The DX 56 DME contains the transmitter, receiver, power supply, and range and speed circuitry.

In the **discussion**, describe each of the mechanism's components and how the mechanism works—its process. Focus on details, word usage, and an organizational pattern that helps your audience easily understand the text.

- **Details.** To develop your discussion, include the following details:

Weight	Density	Make/model
Size (dimensions)	Parts	Texture
Color	Materials (composition)	Capacity
Shape	Identifying numbers	Procedural steps

- **Word usage.** Your word usage in the technical description and process analysis depends on your purpose. For factual, objective technical descriptions or process analyses, use photographic words. For subjective, sales-oriented descriptions or process analyses, use impressionistic words. Photographic words are denotative, quantifiable, and specific. Impressionistic words are vague and connotative. Table 1 shows the difference between photographic and impressionistic words.

TABLE 1 Photographic versus Impressionistic Word Usage

Photographic	Impressionistic
6'9"	tall
350 lb	heavy
gold	precious metal
6,000 shares of United Can	major holdings
700 lumens	bright
0.030 mm	thin
1966 XKE Jaguar	impressive classic car

- **Internal organization.** When describing your topic in the discussion portion of the technical description or process analysis, itemize the topic in some logical sequence. For a technical description, use spatial organization. When a topic is spatially organized, you literally lay out the components as they exist in space. You describe the components as they are seen either from left to right, right to left, top to bottom, bottom to top, inside to outside, or outside to inside. In contrast, when writing your process analysis, you will use chronological organization to show how the product or service works in sequence.

Your **conclusion** depends on your purpose in describing the topic or analyzing how the process works. Some options are as follows:

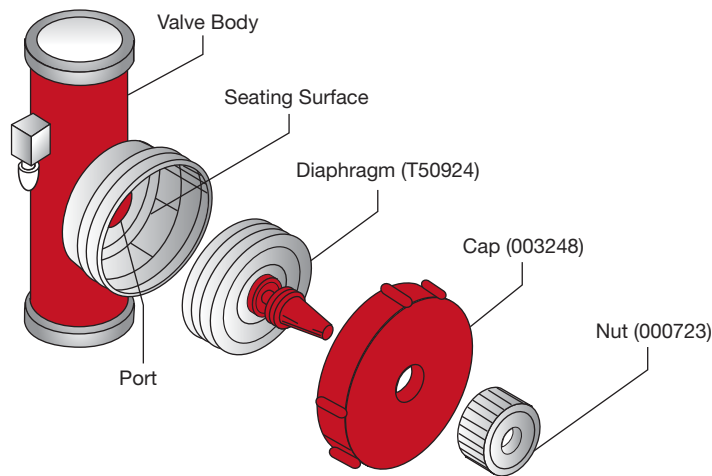
Sales —"Implementation of this product will provide you and your company . . ."	Guarantees —"The product carries a 15-year warranty against parts and labor."	Comparison/contrast —"Compared to our largest competitor, this product is three times more weather resistant." . . .
Uses —"After implementation, you will be able to use this product to . . ."	Testimony —"Our satisfied customers include . . ."	Reiteration of introductory comments —"Thus, the product is composed of the above interchangeable parts and works according to the process explained."

Highlighting Techniques

To aid understanding and allow for easy access, use highlighting techniques, including headings, itemization, and graphics.

You can use line drawings, schematics, CAD/CAM drawings, photographs, architectural renderings, clip art, exploded views, or sectional cutaway views of your topic, each accompanied by *callouts* (labels identifying key components of the mechanism). An example of an exploded view with callouts can be seen in Figure 4.

FIGURE 4 Exhalation Valve with Labeled Callouts



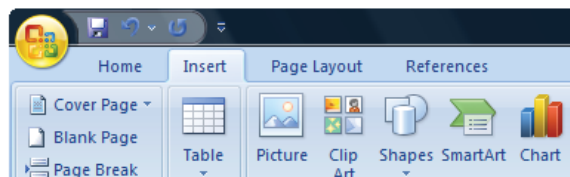
Source: Courtesy of Nelcor Puritan Bennett Corp.

TECHNOLOGY TIPS

How to Make Callouts Using Microsoft Office Word 2010

Callouts are a great way to label mechanism components in your technical descriptions or process analyses. Follow these steps to create callouts.

1. Click **Insert** on the Menu bar.



2. Click **Shapes**. You will see a dropdown box.



(Continued)

TECHNOLOGY TIPS (Continued)

3. Select the type of callout you want to use, such as the **Rounded Rectangular Callout** as shown.



4. When you choose the callout, a crosshair will appear. Use it to draw your callout, as shown.



Note: When you draw the callout, Word opens up the **Format** tab, as shown.



From the **Format** tab, you can change the color of the text box (from **Text Box Styles**), change its line type and density (from **Shape Outline**), create shadow effects, 3-D effects, and add content.

TECHNICAL DESCRIPTIONS AND PROCESS ANALYSIS CHECKLIST

- | | |
|--|---|
| <p>_____ 1. Does the technical description or process analysis have a title noting your topic's name (and any identifying numbers)?</p> <p>_____ 2. Does the technical description or process analysis's introduction (a) state the topic, (b) mention its functions or the purpose, and (c) list the components?</p> <p>_____ 3. Does the technical description or process analysis's discussion use headings to itemize the components for reader-friendly ease of access?</p> <p>_____ 4. Do you need to define the mechanism and its main parts?</p> <p>_____ 5. Does the discussion portion of the description or process analysis specify the following?</p> | <p>_____ 6. Are all of the calculations, measurements, or process steps correct?</p> <p>_____ 7. Do you sum up your discussion using any of the optional conclusions discussed in this chapter?</p> <p>_____ 8. Does your technical description or process analysis provide graphics that are correctly labeled, appropriately placed, and appropriately sized?</p> <p>_____ 9. Do you write clearly and concisely, using a personalized tone?</p> <p>_____ 10. Have you avoided biased language and grammatical and mechanical errors?</p> |
|--|---|
-
- | | |
|-----------|---------------------|
| Colors | Capacities |
| Sizes | Textures |
| Materials | Identifying numbers |

The Writing Process at Work

The following student-written technical description shows how this student used the writing process to create his document.

Prewriting

We asked students to describe a piece of equipment, tool, or mechanism of their choice. One student chose to describe a cash register pole display. To do so, he first had to gather data using brainstorming/listing. He provided the list shown in Figure 5.

FIGURE 5 Prewriting List

Pole PCB • length—15 mm • width—5.1 mm • tube length—10.8 mm • face plate width—2.3 mm • thick—1.7 mm • PCB thickness—0.2 mm • 10 inch stranded wire with female connectors • fiberglass and copper construction Pole Case Assembly • long—15.5 mm • bottom width—2.5 mm • top width—0.9 mm • mounting pole—5 mm high $\times$ 3.2 mm diameter • tongue for mounting—3.1 mm • lower mounting tongue—1.5 mm • side mounting tongue—0.8 mm high • high—6.1 mm • almond-colored plastic Filter • long—15.6 mm • high—6.2 mm • thick—0.7 mm • plastic, blue

Writing

After the prewriting activity, we asked students to draft a technical description. Focusing on overall organization, highlighting, detail, and a hand-drawn graphic, this student wrote a rough draft. He had peer group evaluators suggest revisions (Figure 6).

Rewriting

The student incorporated these suggestions and prepared the finished copy (Figure 7).

FIGURE 6 Student's Rough Draft of a Pole Display with Suggested Revisions from Peer Group Review

No Title *define* *awkward*

The QL169 Customer Pole Display provides the viewing of ~~all transaction data~~ for the customer. The display consist of a printed circuit board, a case assembly, and a filter display. (SP)

Display Circuit Board

- length—15 mm
- width—5.1 mm
- tube length—10.8 mm
- face plate width—2.3 mm
- ~~thick~~—1.7 mm *thickness*
- PCB thickness—0.2 mm
- 10 inch stranded wire with female connectors
- fiberglass and copper construction

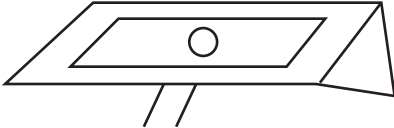
Make All Caps *Number your components*



Display Case Assembly

- ~~long~~—15.5 mm *length*
- bottom width—2.5 mm
- top width—0.9 mm
- mounting pole—5 mm high × 3.2 mm diameter
- ~~tounge~~ for mounting—3.1 mm
- (SP) lower mounting ~~tounge~~—1.5 mm wide
- side mounting ~~tounge~~—0.8 mm high
- ~~high~~—6.1 mm *height*
- almond-colored plastic *vague*

Too vague (specify) *Number your components*



Display Filter

- ~~long~~—15.6 mm *length*
- ~~high~~—6.2 mm *height*
- ~~thick~~—0.7 mm *thickness*
- plastic, blue *what*

Add a conclusion

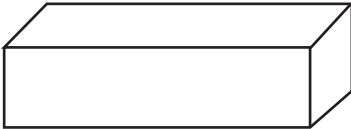


FIGURE 7 Student's Revised Technical Description

Include a definition in the introduction.

Exploded graphics help readers visualize the topic.

QL169 Pole Display

The QL169 pole is an electronic mechanism that provides an alphanumeric display for customer viewing of cash register sales. The display consists of a printed circuit board (PCB) assembly, a case assembly, and a display filter.

Item Description

1. PCB ASSEMBLY
2. CASE ASSEMBLY
3. DISPLAY FILTER

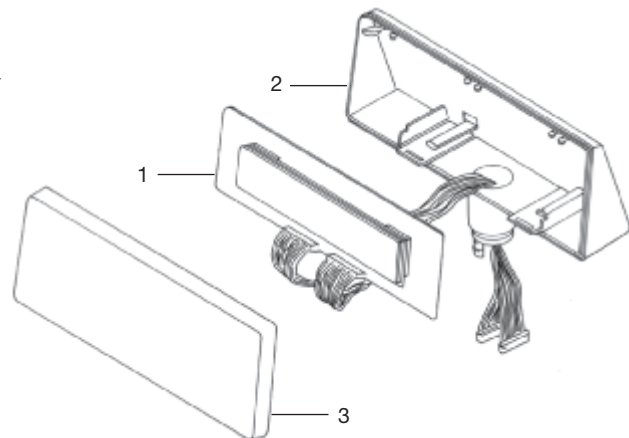


Figure 1. QL169

1.0 PCB Assembly

The printed circuit board, containing the display's electrical circuitry, is constructed of fiberglass with copper etchings.

The board consists of the following features:

- 1.1 Length—15 mm
- 1.2 Width—5.1 mm
- 1.3 Tube length—10.8 mm
- 1.4 Tube faceplate width—2.3 mm
- 1.5 Tube total width—2.8 mm
- 1.6 Tube thickness—1.7 mm
- 1.7 PCB thickness—0.2 mm
- 1.8 20 conductor 10" 22-gauge stranded wire with two AECC female connectors (AECC part #7214-001)
- 1.9 American Display Company blue phosphor display tube (ADC part #1172177)

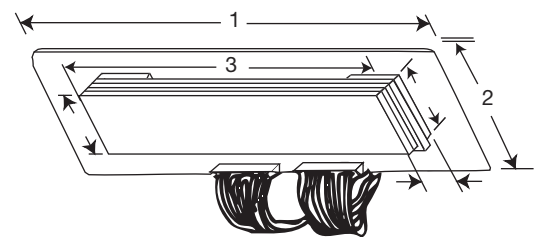


Figure 2. PCB

Specificity of detail adds clarity.

FIGURE 7 (Continued)

2.0 Case Assembly

Almond-colored ABS plastic, used to construct the case assembly, protects the PCB.

2.1 Length—15.5 mm

2.2 Bottom width—2.5 mm

2.3 Top width—0.9 mm

2.4 Mounting pole—5 mm high and 3.2 mm in diameter

2.5 Mounting tongue inside width—3.1 mm from side of assembly

2.6 Lower mounting tongue—1.5 mm wide

2.7 Side mounting tongue—0.8 mm high

2.8 Tongue thickness—0.2 mm

2.9 Height—6.1 mm

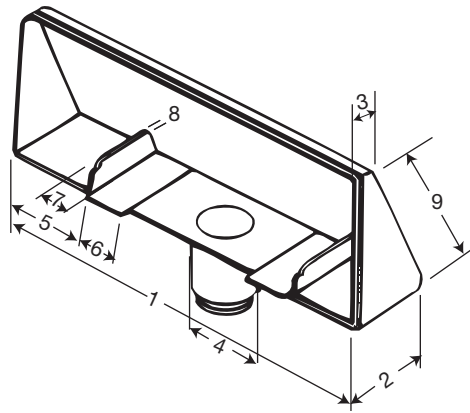


Figure 3. Case Assembly

3.0 Display Filter

3.1 Length—15.6 mm

3.2 Height—6.2 mm

3.3 Thickness—0.7 mm

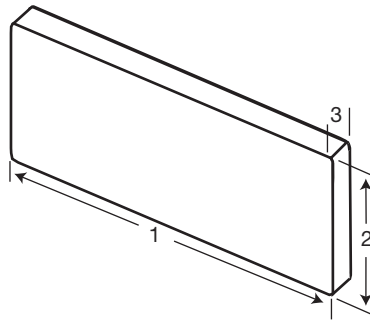


Figure 4. Display Filter

Transparent blue plastic, used to construct the display filter, allows the customer to view the readings. The QL169 pole provides easy viewing of clerk transactions and ensures cashier accuracy.

The conclusion focuses on the end-user benefits.

CHAPTER HIGHLIGHTS

1. Technical descriptions are often used in user manuals, product specifications, construction designs, reports, and sales literature.
2. Process analyses tell how a tool, mechanism, piece of equipment, product, or service works.
3. The introduction of a technical description or process analysis specifies and defines the topic; explains the topic's functions, capabilities, or processes; and might include a list of components, parts, or equipment.
4. Graphics with callouts make a technical description and process analysis easier to understand for the reader.
5. Follow a spatial pattern of organization in a technical description.
6. Follow a chronological order of organization when describing a process.
7. Word usage for descriptions and processes will be photographic and/or impressionistic.
8. Use headings and subheadings to organize the content of your description or process analysis.
9. Use line drawings, schematics, CAD/CAM drawings, photographs, architectural renderings, clip art, exploded views, or sectional cutaway views of your topic for the technical description or process analysis.
10. Preface your text with a title precisely stating the topic. This could be the name of the mechanism, tool, piece of equipment, landform, product, or service you are describing or analyzing.

APPLY YOUR KNOWLEDGE



CASE STUDY

Nitrous Systems Biotechnology, Inc. plans to expand its company. Part of this expansion includes the need to hire 50 new biotechnicians. Each will need a tablet PC. But what kind of tablet PC best meets the biotechnicians' needs: one with a touch screen or a pen-enabled interface requiring a digital stylus? You have been asked to research tablet PCs. Your manager wants a brief comparison/contrast, complete with a product description and a process analysis explaining the differences between touch screens and pen-enabled interfaces requiring a digital stylus. What are the technical specifications for each interface, and how does each interface work? Research this topic and write the technical description, complete with process analysis, to meet your manager's request.

INDIVIDUAL AND TEAM PROJECTS

1. Write a technical description, either individually or as a team. To do so, first select a topic. You can describe any tool, mechanism, or piece of equipment. However, don't choose a topic too large to describe accurately. To provide a thorough and precise description, you will need to be exact and minutely detailed. A large topic, such as a computer, an oscilloscope, a respirator, or a Boeing airliner, would be too demanding for a two- to four-page description. On the other hand, do not choose a topic that is too small, such as a paper clip, a nail, or a shoestring. Choose a topic that provides you with a challenge but that is manageable. You might write about any of the following topics:

USB Flash Drive	Backpack	DVD remote control
Wrench	Computer mouse	Mechanical pencil
Screwdriver	Light bulb	Ballpoint pen
Pliers	Calculator	Computer monitor
Wall outlet	Automobile tire	Smartphone

2. Write a process analysis, either individually or as a team, on one of the following topics:
 - How does blood coagulate?
 - How does an X-ray machine work?
 - How do Wi-Fi connections work?
 - How do rotary engines work?
 - How do fuel gauges work?
 - How is metal welded?
 - How does a metal detector work?
 - How do browsers work?
 - How do computer viruses work?
 - How does a wireless mouse work?
3. Select a simple topic for description, such as a pencil, coffee cup, toothbrush, or textbook (you can use brainstorming/listing to come up with additional topics). Describe this item without mentioning what it is or providing any graphics. Then, read your description to a group of students/peers and ask them to draw what you have described. If your verbal description is good, their drawings will resemble your topic. If their drawings are off base, you haven't succeeded in providing an effective description.
4. In degree-specific teams, choose a topic from your area of interest and expertise to describe (including a process analysis). For example, students majoring in HVAC could describe a humidifier and explain how it works. Biotech and nursing students could describe and provide a process analysis for a nebulizer, blood pressure monitor, or glucometer. EMT students could write about defibrillators. Automotive technicians could describe and include a process analysis for jumper cables, battery chargers, or hydraulic jacks. Welding students could write about MIG, TIG, or Stick welding.

PROBLEM-SOLVING THINK PIECE

Read the following process analysis and reorganize the numbered sentences to achieve a clear, chronological order. If necessary, research this process to learn more about how blood clots.

How Does Blood Clot?

1. In our bodies, blood can clot due to platelets and the thrombin system.
2. When bleeding occurs, chemical reactions make the surface of the platelet “sticky.” These sticky platelets adhere to the wall of blood vessels where bleeding has occurred.
3. Platelets, tiny cells created in our bone marrow, travel in the bloodstream and wait for a bleeding problem.
4. Soon, a “white clot” is formed, so called because the clotted platelets look white.
5. Blood clots consist of both platelets and fibrin. The fibrin strands bind the platelets and make the clot stable.
6. In the thrombin system, several blood proteins become active when bleeding occurs. Clotting reactions produce fibrin, long, sticky strings. These sticky strands catch red blood cells and form a “red clot.”
7. Primarily, arteries clot due to platelets, while veins clot due to our thrombin system.

WEB WORKSHOP

1. You are ready to purchase a product. This could include printers, monitors, smartphones, scanners, PCs, tablet PCs, speakers, cables, adapters, automotive engine hoists, generators, battery chargers, jacks, power tools, truck boxes, screws, bolts, nuts, rivets, hand tools, and more. A great place to shop is online. By going to an online search engine, you can find not only prices for your products but also technical descriptions or technical specifications. These will help you determine if the product has the size, shape, materials, and capacity you are looking for.

Go online to search for a product of your choice and review the technical description or specifications provided. Using the criteria in this chapter and your knowledge of effective technical communication techniques, analyze your findings.

- Are graphics used to help you visualize the product?
 - Are callouts used to help you identify parts of the product?
 - Are high-tech terms defined?
 - Is the use of the product explained?
 - a. Report your findings, either in an oral presentation or in writing (e-mail message, memo, letter, or report).
 - b. Rewrite any of the technical descriptions that need improvement according to the criteria provided in this textbook.
2. HowStuffWorks (<http://www.howstuffworks.com/index.htm>) is an extensive online library of process analyses. Revise a process analysis in this Web site.

GRAMMAR, PUNCTUATION, MECHANICS, AND SPELLING

Correct organization and development of your memos, letters, or reports is important for the success of your technical communication. However, no one will be impressed with the quality of your work, or with you, if your writing is riddled with errors in sentence construction or punctuation. Your written correspondence is often your first contact with business associates. Many people mistakenly believe that only English teachers notice grammatical errors and wield red pens, but businesspeople as well take note of such errors and may see the writer as less competent.

We were working recently with a young executive who is employed by a branch of the federal government. This executive told us that whenever his supervisor found a spelling error in a subordinate's report, this report was paraded around the office. Everyone was shown the mistake and had a good laugh over it, and the report was then returned to the writer for correction. Our acquaintance assured us that all of this was in good-natured fun. However, he also said that employees quickly learned to edit and proofread their written communication to avoid such public displays of their errors.

Your writing at work may not be exposed to such scrutiny by coworkers. Instead, your writing may go directly to another firm, and those readers will see your mistakes. To avoid this problem, you must evaluate your writing for grammar, punctuation, and spelling errors. If you don't, your customers, bosses, and colleagues will.

Grammar Rules

To understand the fundamentals of grammar, you must first understand the basic components of a sentence.

A correctly constructed sentence consists of a subject and a predicate (some sentences also include a phrase or phrases).

EXAMPLE ►

The meeting	began	at 4:00 A.M.
subject	predicate	phrase

Subject: The *doer* of the action; the subject usually precedes the predicate.

Predicate: The *action* in the sentence

EXAMPLE ►

He	ran	to the office to avoid being late.
doer	action	phrases

If the subject and the predicate (a) express a complete thought and (b) can stand alone, you have an *independent clause*.

EXAMPLE ►

The meeting began	at 4:00 P.M.
independent clause	phrase

A *phrase* is a group of related words that does not contain a subject and a predicate and cannot stand alone or be punctuated as a sentence. The following are examples of phrases:

at the house
in the box
on the job
during the interview

If a clause is dependent, it cannot stand alone.

◀ EXAMPLE

Although he tried to hurry, he was late for the meeting.
dependent clause independent clause

◀ EXAMPLE

He was late for the meeting although he tried to hurry.
independent clause dependent clause

NOTE: When a dependent clause begins a sentence, use a comma before the independent clause. However, when an independent clause begins a sentence, do not place a comma before the dependent clause.

Agreement between Pronoun and Antecedent (Referent)

A pronoun has to agree in gender and number with its antecedent.

Susan went on *her* vacation yesterday.
The *people* who quit said that *they* deserved raises.

Problems often arise when a singular indefinite pronoun is the antecedent. The following antecedents require singular pronouns: *anybody*, *each*, *everybody*, *everyone*, *somebody*, and *someone*.

incorrect

Anyone can pick up *their* applications at the job placement center.

correct

Anyone can pick up *his* or *her* application at the job placement center.

Problems also arise when the antecedent is separated from the pronoun by numerous words.

incorrect

Even when the best *employee* is considered for a raise, *they* often do not receive it.

correct

Even when the best *employee* is considered for a raise, *he* or *she* often does not receive it.

Agreement between Subject and Verb

Writers sometimes create disagreement between subjects and verbs, especially if other words separate the subject from the verb. To ensure agreement, ignore the words that come between the subject and verb.

incorrect

Her *boss* undoubtedly *think* that all the employees want promotions.

correct

Her *boss* undoubtedly *thinks* that all the employees want promotions.

incorrect

The *employees* who sell the most equipment *is* going to Hawaii for a week.

correct

The *employees* who sell the most equipment *are* going to Hawaii for a week.

If a sentence contains two subjects (a compound subject) connected by *and*, use a plural verb.

incorrect

Joe and Tiffany *was* both selected employee of the year.

correct

Joe and Tiffany *were* both selected employees of the year.

incorrect

The bench workers and their supervisor *is* going to work closely to complete this project.

correct

The bench workers and their supervisor *are* going to work closely to complete this project.

Add a final *s* or *es* to create most plural subjects or singular verbs, as follows:

PLURAL SUBJECTS	SINGULAR VERBS
bosses hire	a boss hires
employees demand	an employee demands
experiments work	an experiment works
attitudes change	the attitude changes

If a sentence has two subjects connected by *either . . . or*, *neither . . . nor*, or *not only . . . but also*, the verb should agree with the closest subject. This also makes the sentence less awkward.

EXAMPLE ►

Either the salespeople or the warehouse worker *deserves* raises.
 Not only the warehouse worker but also the salespeople *deserve* raises.
 Neither the salespeople nor the warehouse worker *deserves* raises.

Singular verbs are used after most indefinite pronouns such as the following:

another	everything
anybody	neither
anyone	nobody
anything	no one
each	nothing
either	somebody
everybody	someone
everyone	something

Anyone who works here *is* guaranteed maternity leave.

Everybody wants the company to declare a profit this quarter.

Singular verbs often follow collective nouns such as the following:

class	organization
corporation	platoon
department	staff
group	team

The *staff* *is* sending the boss a bouquet of roses.

Comma Splice

A *comma splice* occurs when two independent clauses are joined by a comma rather than separated by a period or semicolon.

incorrect

Sue was an excellent employee, she got a promotion.

Several remedies will correct this error.

1. Separate the two independent clauses with a semicolon.

correct

Sue was an excellent employee; she got a promotion.

2. Separate the two independent clauses with a period.

correct

Sue was an excellent employee. She got a promotion.

3. Separate the two independent clauses with a comma and a *coordinating conjunction* (*and, but, or, for, so, yet*).

correct

Sue was an excellent employee, *so* she got a promotion.

4. Separate the two independent clauses with a semicolon (or a period), a conjunctive adverb, and a comma. *Conjunctive adverbs* include *also*, *additionally*, *consequently*, *furthermore*, *however*, *instead*, *moreover*, *nevertheless*, *therefore*, and *thus*.

correct

Sue was an excellent employee; *therefore*, she got a promotion.

or

Sue was an excellent employee. *Therefore*, she got a promotion.

5. Use a *subordinating conjunction* to make one of the independent clauses into a dependent clause. Subordinating conjunctions include *after*, *although*, *as*, *because*, *before*, *even though*, *if*, *once*, *since*, *so that*, *though*, *unless*, *until*, *when*, *where*, and *whether*.

correct

Because Sue was an excellent employee, she got a promotion.

Faulty or Vague Pronoun Reference

A pronoun must refer to a specific noun (its antecedent). Problems arise when (a) there is an excessive number of pronouns (causing vague pronoun reference) and (b) there is no specific noun as an antecedent. Notice that there seems to be an excessive number of pronouns in the following passage, and the antecedents are unclear.

EXAMPLE ►

Although Bob had been hired over two years ago, *he* found that *his* boss did not approve *his* raise. In fact, *he* was also passed over for *his* promotion. The boss appears to have concluded that *he* had not exhibited zeal in *his* endeavors for their business. Instead of being a highly valued employee, *he* was not viewed with pleasure by those in authority. Perhaps it would be best if *he* considered *his* options and moved to some other company where *he* might be considered in a new light.

The excessive and vague use of *he* and *his* causes problems for readers. Do these words refer to Bob or to his boss? You are never completely sure. To avoid this problem, limit pronoun usage, as in the following revision.

EXAMPLE ►

Although Bob had been hired over two years ago, he found that his boss, Joe, did not approve his raise. In fact, Bob was also passed over for promotion. Joe appears to have concluded that Bob had not exhibited zeal in his endeavors for their business. Instead of being a highly valued employee, Bob was not viewed with pleasure by those in authority. Perhaps it would be best if Bob considered his options and moved to some other company where he might be considered in a new light.

To make the preceding paragraph more precise, we have replaced vague pronouns (*he* and *his*) with exact names (*Bob* and *Joe*).

Fragments

A *fragment* occurs when a group of words is incorrectly used as an independent clause. Often the group of words begins with a capital letter and has end punctuation but is missing either a subject or a predicate.

incorrect

Working with computers.
(lacks a predicate and does not express a complete thought)

The group of words may have a subject and a predicate but be a dependent clause.

incorrect

Although he enjoyed working with computers.
(has a subject, *he*, and a predicate, *enjoyed*, but is a dependent clause because it is introduced by the subordinate conjunction *although*)

It is easy to remedy a fragment by doing one of the following:

- Add a subject.
- Add a predicate.
- Add both a subject and a predicate.
- Add an independent clause to a dependent clause.

correct

Joe found that working with computers used his training.
(subject, *Joe*, and predicate, *found*, have been added)

correct

Although he enjoyed working with computers, he could not find a job in a computer-related field.
(independent clause, *he could not find a job*, added to the dependent clause, *Although he enjoyed working with computers*)

Fused Sentence

A *fused sentence* occurs when two independent clauses are connected with no punctuation.

incorrect

The company performed well last quarter its stock rose several points.

There are several ways to correct this error.

1. Write two sentences separated by a period.

correct

The company performed well last quarter. Its stock rose several points.

2. Use a comma and a coordinating conjunction to separate the two independent clauses.

correct

The company performed well last quarter, *so* its stock rose several points.

3. Use a subordinating conjunction to create a dependent clause.

correct

Because the company performed well last quarter, its stock rose several points.

4. Use a semicolon to separate the two independent clauses.

correct

The company performed well last quarter; its stock rose several points.

5. Separate the two independent clauses with a semicolon, a conjunctive adverb or a transitional word or phrase, and a comma.

correct

The company performed well last quarter; *therefore*, its stock rose several points.

correct

The company performed well last quarter; *for example*, its stock rose several points.

The following are transitional words and phrases, listed according to their use.

TO ADD

again	in addition
also	moreover
besides	next
first	second
furthermore	still

TO COMPARE/CONTRAST

also	nevertheless
but	on the contrary
conversely	still
in contrast	

TO PROVIDE EXAMPLES

for example	of course
for instance	put another way
in fact	to illustrate

TO SHOW PLACE

above	here
adjacent to	nearby
below	on the other side
elsewhere	there
further on	

To Reveal Time

afterward	second
first	shortly
meanwhile	subsequently
presently	thereafter

To Summarize

all in all	last
finally	on the whole
in conclusion	therefore
in summary	thus

Modification

A *modifier* is a word, phrase, or clause that explains or adds details about other words, phrases, or clauses.

Misplaced Modifiers. A *misplaced modifier* is one that is not placed next to the word it modifies.

incorrect

He had a heart attack *almost* every time he was reviewed by his supervisor.

correct

He almost had a heart attack every time he was reviewed by his supervisor.

incorrect

The worker had to *frequently* miss work.

correct

The worker frequently had to miss work.

Dangling Modifiers. A *dangling modifier* is a modifier that is not placed next to the word or phrase it modifies. To avoid confusing your readers, place modifiers next to the word(s) they refer to. Don't expect your readers to guess at your meaning.

incorrect

While working, tiredness overcame them.
(Who was working? Who was overcome by tiredness? *While working* should modify *staff*, but the word *staff* does not appear in the sentence.)

correct

While working, the staff became tired.

incorrect

After soldering for two hours, the equipment was ready for shipping. (Who had been soldering for two hours? Not the equipment!)

correct

After soldering for two hours, the technicians prepared the equipment for shipping.

Parallelism

All items in a list should be parallel in grammatical form. Avoid mixing phrases and sentences (independent clauses).

incorrect

We will discuss the following at the department meeting:

1. Entering mileage in logs (phrase)
2. All employees have to enroll in a training seminar. (sentence)
3. Purpose of quarterly reviews (phrase)
4. Some data processors will travel to job sites. (sentence)

correct

We will discuss the following at the department meeting:

- | | | |
|--|---|---------|
| <ol style="list-style-type: none"> 1. Entering mileage in logs 2. Enrolling in training seminars 3. Reviewing employee performance quarterly 4. Traveling to job sites | } | phrases |
|--|---|---------|

correct

At the department meeting, you will learn how to do the following:

- | | | |
|--|---|-----------|
| <ol style="list-style-type: none"> 1. Enter mileage in logs. 2. Enroll in training seminars. 3. Review employee performance quarterly. 4. Travel to job sites. | } | sentences |
|--|---|-----------|

Punctuation

Apostrophe (')

Place an *apostrophe* before the final *s* in a singular word to indicate possession.

EXAMPLE ►

Jim's tool chest is next to the furnace.

Place the apostrophe after the final *s* if the word is plural.

EXAMPLE ►

The employees' reception will be held next week.

Don't use an apostrophe to make singular abbreviations plural.

incorrect

The EXT's will be shipped today.

correct

The EXTs will be shipped today.

Colon (:

Use a *colon* after a salutation.

Dear Mr. Harken:

◀ EXAMPLE

In addition, use a colon after an emphatic or cautionary word if explanations follow.

Note: Hand-tighten the nuts.

Caution: Wash thoroughly if any mixture touches your skin.

◀ EXAMPLE

Finally, use a colon after an independent clause to precede a quotation, list, or example.

She said the following: “No comment.”

These supplies for the experiment are on order: a plastic hose, two batteries, and several chemicals.

The problem has two possible solutions: Hire four more workers, or give everyone a raise.

◀ EXAMPLE

NOTE: In the preceding examples, the colon follows an independent clause.

A common mistake is to place a colon after an incomplete sentence. Except for salutations and cautionary notes, whatever precedes a colon *must* be an independent clause.

incorrect

The two keys to success are: earning money and spending wisely.

correct

The two keys to success are earning money and spending wisely.

or

The two keys to success are as follows: earning money and spending wisely.

or

The two keys to success are as follows:

1. Earning money
2. Spending wisely

Comma (,)

Writers often get in trouble with *commas* when they employ one of two common words of wisdom.

- When in doubt, leave it out.
- Use a comma when there is a pause.

Both rules are inexact. Writers use the first rule to justify the complete avoidance of commas; they use the second rule to sprinkle commas randomly throughout their writing. On the contrary, commas have several specific conventions that determine usage.

1. Place a comma before a coordinating conjunction (*and, but, or, for, so, yet*) linking two independent clauses.

EXAMPLE ►

You are the best person for the job, *so* I will hire you.
We spent several hours discussing solutions to the problem, *but* we failed to decide on a course of action.

2. Use commas to set off introductory comments.

EXAMPLE ►

First, she soldered the components.
In business, people often have to work long hours.
To work well, you need to get along with your coworkers.
If you want to test equipment, do so by 5:25 P.M.

3. Use commas to set off sentence interrupters.

EXAMPLE ►

The company, started by my father, did not survive the last recession.
Mrs. Patel, the proprietor of the store, purchased a wide array of merchandise.

4. Set off parenthetical expressions with commas.

EXAMPLE ►

A worker, it seems, should be willing to try new techniques.
The highway, by the way, needs repairs.

5. Use commas after each item in a series of three or more.

EXAMPLE ►

Prakash, Mirren, and Justin were chosen as employees of the year.
We found the following problems: corrosion, excessive machinery breakdowns, and power failures.

6. Use commas for long numbers.

EXAMPLE ►

She earns \$100,000 before taxes.

NOTE: Very large numbers are often written as words.

EXAMPLE ►

Our business netted over \$2 million in 2014.

7. Use commas to separate the day and year when they are part of a sentence.

EXAMPLE ►

The company hired her on September 7, 2014, to be its bookkeeper.

NOTE: If the year is used as an adjective, do not follow it with a comma.

EXAMPLE ►

The 2014 corporate report came out today.

8. Use commas to separate the city from the state and the state from the rest of the sentence.

The new warehouse in Austin, Texas, will promote increased revenues.

◀ EXAMPLE

NOTE: If you omit either the city or the state, you do not need commas.

The new warehouse in Austin will promote increased revenues.

◀ EXAMPLE

Dash (—)

A *dash*, typed as two consecutive hyphens with no spaces before or after, is a versatile punctuation mark. Most word processing programs now include an em dash character. Use dashes in the following ways.

1. After a heading and before an explanation.

Forecasting—Joe and Joan will be in charge of researching fourth-quarter production quotas.

◀ EXAMPLE

2. To indicate an emphatic pause.

You will be fired—unless you obey company rules.

◀ EXAMPLE

3. To highlight a new idea.

Here's what we can do to improve production quality—provide on-the-job training, salary incentives, and quality controls.

◀ EXAMPLE

4. Before and after an explanatory or appositive series.

Three people—Sue, Luci, and Tom—are essential to the smooth functioning of our office.

◀ EXAMPLE

Ellipses (...)

Ellipses (three spaced periods) indicate omission of words within quoted materials.

“Six years ago, prior to incorporating, the company had to pay extremely high federal taxes.”

“Six years ago, . . . the company had to pay extremely high federal taxes.”

◀ EXAMPLE

Exclamation Point (!)

Use an *exclamation point* after strong statements, commands, or interjections.

You must work harder!
Do not use the machine!
Danger!

◀ EXAMPLE

Hyphen (-)

A *hyphen* is used in the following ways.

1. To indicate the division of a word at the end of a typed line. Remember, this division must occur between syllables.
2. To create a compound adjective.

EXAMPLE ►

He is a well-known engineer.
Until her death in 2014, she was a world-renowned chemist.
Tom is a 24-hour-a-day student.

3. To join the numerator and denominator of fractions.

EXAMPLE ►

Four-fifths of the company want to initiate profit sharing.

4. To write out two-word numbers.

EXAMPLE ►

Twenty-six people attended the conference.

Parentheses ()

Parentheses enclose abbreviations, numbers, words, or sentences for the following reasons.

1. To define a term or provide an abbreviation for later use.

EXAMPLE ►

We belong to the Society for Technical Communication (STC).

2. To clarify preceding information in a sentence.

EXAMPLE ►

The people in attendance (all regional sales managers) were proud of their accomplishments.

3. To number items in a series.

EXAMPLE ►

The company should initiate (1) new personnel practices, (2) a probationary review board, and (3) biannual raises.

Period (.)

A *period* must end a declarative sentence (independent clause).

EXAMPLE ►

I found the business trip rewarding.

Periods are often used with abbreviations.

D.C.	Mrs.	A.M. or a.m.
e.g.	Ms.	P.M. or p.m.
Mr.		

It is incorrect to use periods with abbreviations for organizations and associations.

incorrect

S.T.C. (Society for Technical Communication)

correct

STC (Society for Technical Communication)

State abbreviations do not require periods if you use two capital letters. (Continue to use a period after a capital letter and a lowercase letter such as Tx., Ks., or Mo.)

incorrect

KS. (Kansas)
MO. (Missouri)
TX. (Texas)

correct

KS
MO
TX

Question Mark (?)

Use a *question mark* after direct questions.

Do the lab results support your theory?
Will you work at the main office or at the branch?

◀ **EXAMPLE**

Quotation Marks (" ")

Quotation marks are used in the following ways.

1. When citing direct quotations.

He said, "Your division sold the most compressors last year."

◀ **EXAMPLE**

NOTE: When you are citing a quotation within a quotation, use double quotation marks (" ") and single quotation marks (' ').

Kim's supervisor, quoting the CEO, said the following to explain the new policy regarding raises: "Only employees who deserve them will receive merit raises."

◀ **EXAMPLE**

2. To note the title of an article or a subdivision of a report.

EXAMPLE ►

The article “Robotics in Industry Today” was an excellent choice as the basis of your speech.
Section III, “Waste Water in District 9,” is pertinent to our discussion.

When using quotation marks, abide by the following punctuation conventions:

- Commas and periods always go *inside* quotation marks.

EXAMPLE ►

She said, “Our percentages are fixed.”

- Colons and semicolons always go *outside* quotation marks.

EXAMPLE ►

He said, “The supervisor hasn’t decided yet”; however, he added that the decision would be made soon.

- Exclamation points and question marks go inside the quotation marks if the quoted material is either exclamatory or a question. However, if the quoted material is not exclamatory or a question, then these punctuation marks go outside the quotation marks.

EXAMPLE ►

John said, “Don’t touch that liquid. It’s boiling!”

(Although the sentence isn’t exclamatory, the quotation is. Thus, the exclamation point goes inside the quotation marks.)

EXAMPLE ►

How could she say, “We haven’t purchased the equipment yet”?

(Although the quotation isn’t a question, the sentence is. Thus, the question mark goes outside the quotation marks.)

Semicolon (;)

Semicolons are used in the following instances.

1. Between two independent clauses *not* joined by a coordinating conjunction.

EXAMPLE ►

The light source was unusual; it emanated from a crack in the plastic surrounding the cathode.

2. To separate items in a series containing internal commas.

EXAMPLE ►

When the meeting was called to order, all members were present, including Susan Johnson, the president; Jewel Smith, the vice president; Harold Holbert, the treasurer; and Linda Hamilton, the secretary.

Mechanics

Abbreviations

Never use an abbreviation that your reader will not understand. A key to clear technical writing is to write on a level appropriate to your reader. You may use the following familiar abbreviations without explanation: *Mrs.*, *Dr.*, *Mr.*, *Ms.*, and *Jr.*

A common mistake is to abbreviate inappropriately. For example, some writers abbreviate *and* as follows:

I quit my job & planned to retire young.

◀ **EXAMPLE**

This is too colloquial for professional technical writing. Spell out *and* when you write.

The majority of abbreviation errors occur when writers incorrectly abbreviate states and technical terms.

States. Writers often abbreviate the names of states incorrectly. Use the U.S. Postal Service abbreviations in addresses.

Abbreviations for States

AL	Alabama	MT	Montana
AK	Alaska	NC	North Carolina
AZ	Arizona	ND	North Dakota
AR	Arkansas	NE	Nebraska
CA	California	NV	Nevada
CO	Colorado	NH	New Hampshire
CT	Connecticut	NJ	New Jersey
DE	Delaware	NM	New Mexico
FL	Florida	NY	New York
GA	Georgia	OH	Ohio
HI	Hawaii	OK	Oklahoma
IN	Indiana	OR	Oregon
IA	Iowa	PA	Pennsylvania
ID	Idaho	RI	Rhode Island
IL	Illinois	SC	South Carolina
KS	Kansas	SD	South Dakota
KY	Kentucky	TN	Tennessee
LA	Louisiana	TX	Texas
ME	Maine	UT	Utah
MD	Maryland	VT	Vermont
MA	Massachusetts	VA	Virginia
MI	Michigan	WA	Washington
MN	Minnesota	WV	West Virginia
MS	Mississippi	WI	Wisconsin
MO	Missouri	WY	Wyoming

Technical Terms. Units of measurement and scientific terms must be abbreviated accurately to ensure that they will be understood. Writers often use such abbreviations incorrectly. For example, “The unit measured 7.9 cent.” is inaccurate. The correct abbreviation for centimeter is *cm*, not *cent*. Use the following abbreviation conventions.

Technical Abbreviations for Units of Measurement and Scientific Terms

absolute	abs	current (electric)	I
alternating current	AC	cycles per second	CPS
American wire gauge	AWG	decibel	dB
ampere	amp	decigram	dg
ampere-hour	amp-hr	deciliter	dL
amplitude modulation	AM	decimeter	dm
angstrom unit	Å	degree	deg
atmosphere	atm	dekagram	dkg
atomic weight	at wt	dekaliter	dL
audio frequency	AF	dekameter	dkm
azimuth	az	dewpoint	DP
barometer	bar.	diameter	dia
barrel, barrels	bbl	direct current	DC
billion electron volts	BeV	dozen	doz (or dz)
biochemical oxygen demand	BOD	dram	dr
board foot	bdf	electromagnetic force	emf
Brinell hardness number	BHN	electron volt	eV
British thermal unit	Btu	elevation	el (or elev)
bushel	bu	equivalent	equiv
calorie	cal	Fahrenheit	F
candela	cd	farad	F
Celsius	C	faraday	f
center of gravity	cg	feet, foot	ft
centimeter	cm	feet per second	ft/sec
circumference	cir	fluid ounce	floz
cologarithm	colog	foot board measure	fbm
continuous wave	CW	foot-candle	ft-c
cosine	cos	foot-pound	ftlb
cotangent	cot	frequency modulation	FM
cubic centimeter	cc	gallon	gal
cubic foot	cu ft (or ft ³)	gallons per day	GPD
cubic feet per second	cfs	gallons per minute	GPM
cubic inch	cu in. (or in. ³)	grain	gr
cubic meter	cu m (or m ³)	grams	g (or gm)
cubic yard	cu yd (or yd ³)	gravitational acceleration	g

(continued)

TECHNICAL DESCRIPTIONS AND PROCESS ANALYSES

hectare	ha	maximum	max
hectoliter	hl	megacycle	mc
hectometer	hm	megahertz	MHz
henry	H	megawatt	MW
hertz	Hz	meter	m
high frequency	HF	microampere	μamp
horsepower	hp	microinch	μin.
horsepower-hours	hp-hr	microsecond	μsec
hour	hr	microwatt	μw
hundredweight	cwt	miles per gallon	mpg
inch	in.	milliampere	mA
inch-pounds	in.-lb	millibar	mb
infrared	IR	millifarad	mF
inner diameter or inside dimensions	ID	milligram	mg
intermediate frequency	IF	milliliter	ml
international unit	IU	millimeter	mm
joule	J	millivolt	mV
Kelvin	K	milliwatt	mW
kilocalorie	kcal	minute	min
kilocycle	kc	nautical mile	NM
kilocycles per second	kc/sec	negative	neg or -
kilogram	kg	number	no.
kilohertz	kHz	octane	oct
kilojoule	kJ	ounce	oz
kiloliter	kl	outside diameter	OD
kilometer	km	parts per billion	ppb
kilovolt	kV	parts per million	ppm
kilovolt-amperes	kVa	pascal	pas
kilowatt-hours	kWH	positive	pos or +
lambert	L	pound	lb
latitude	lat	pounds per square inch	psi
length	l	pounds per square inch absolute	psia
linear	lin	pounds per square inch gauge	psig
linear foot	linft	quart	qt
liter	l	radio frequency	RF
logarithm	log.	radian	rad
longitude	long.	radius	r
low frequency	LF	resistance	r
lumen	lm	revolution	rev
lumen-hour	lm-hr	revolutions per minute	rpm
		second	s (or sec)

Technical Abbreviations for Units of Measurement and Scientific Terms (continued)

secant	sec	ultra high frequency	UHF
specific gravity	sp gr (or SG)	vacuum	vac
square foot	ft ²	very high frequency	VHF
square inch	in. ²	volt-ampere	VA
square meter	m ²	volt	V
square mile	mi ²	volts per meter	V/m
tablespoon	tbs (or tbsp)	volume	vol
tangent	tan	watt-hour	whr
teaspoon	tsp	watt	W
temperature	t	wavelength	WL
tensile strength	ts	weight	wt
thousand	m	yards	y (or yd)
ton	t	years	y (or yr)

Capital Letters

Capitalize the following:

1. Proper nouns.

EXAMPLE ►

people	cities	countries	companies	schools	buildings
Susan	Houston	Italy	Bendix	Harvard	Oak Park Mall

2. People's titles (only when they precede the name).

EXAMPLE ►

Governor Sally Renfro
or
 Sally Renfro, governor
 Technical Supervisor Todd Blackman
or
 Wes Schneider, the technical supervisor

3. Titles of books, magazines, plays, movies, television programs, and CDs (excluding the prepositions and all articles after the first article in the title).

EXAMPLE ►

Mad Men
The Colbert Report
The Catcher in the Rye
The Twilight Saga: Breaking Dawn
American Idol

4. Names of organizations.

Girl Scouts
Phoenix, AZ, Regional Home Care Association
Kansas City Regional Council for Higher Education
Programs for Technical and Scientific Communication
American Civil Liberties Union
Students for a Democratic Society

◀ EXAMPLE

5. Days of the week, months, and holidays.

Monday
December
Thanksgiving

◀ EXAMPLE

6. Races, religions, and nationalities.

American Indian
Jewish
Polish

◀ EXAMPLE

7. Events or eras in history.

the Gulf War
the Vietnam War
World War II

◀ EXAMPLE

8. North, South, East, and West (when used to indicate geographic locations).

Atlanta is one of the fastest growing cities in the South.

◀ EXAMPLE

NOTE: Don't capitalize these words when giving directions.

We were told to drive south three blocks and then to turn west.

◀ EXAMPLE

9. The first word of a sentence.

10. Don't capitalize any of the following:

Seasons—spring, fall, summer, winter
Names of classes—sophomore, senior
General groups—middle management, infielders, surgeons

◀ EXAMPLE

Numbers

Write out numbers one through nine. Use numerals for numbers 10 and above.

10	12
104	2,093
536	5,550,286

Although the preceding rules cover most situations, there are exceptions.

1. Use numerals for all percentages.

EXAMPLE ►

2 percent	18 percent	25 percent
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2. Use numerals for addresses.

EXAMPLE ►

12 Elm	935 W. Harding
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3. Use numerals for miles per hour.

EXAMPLE ►

5 mph	225 mph
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4. Use numerals for time.

EXAMPLE ►

3:15 A.M.

5. Use numerals for dates.

EXAMPLE ►

May 31, 2014

6. Use numerals for monetary values.

EXAMPLE ►

\$45	\$.95	\$2 million
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7. Use numerals for units of measurement.

EXAMPLE ►

14 feet	6 3/4 inches	16 mm 10 V
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8. Do not use numerals to begin sentences.

incorrect

568 people were fired last August.

correct

Five hundred sixty-eight people were fired last August.

9. Do not mix numerals and words when writing numbers. When two or more numbers appear in a sentence and one of them is 10 or more, figures are used.

We attended 4 meetings over a 16-day period.

◀ EXAMPLE

10. Use numerals and words in a compound number adjective to avoid confusion.

The worker needed six 2-inch nails.

◀ EXAMPLE

Spelling

The following is a list of commonly misspelled or misused words. You can avoid many common spelling errors if you familiarize yourself with these words. Remember to run spell check; also remember that spell check will not understand context. The incorrect word contextually could be spelled correctly.

accept, except	council, counsel	personal, personnel
addition, edition	desert, dessert	principal, principle
access, excess	disburse, disperse	quiet, quite
advise, advice	fiscal, physical	rite, right, write
affect, effect	forth, fourth	stationery, stationary
all ready, already	incite, insight	their, there, they're
assistants, assistance	its, it's	to, too, two
bare, bear	loose, lose	whose, who's
brake, break	miner, minor	your, you're
coarse, course	passed, past	
cite, site, sight	patients, patience	

Apply Your Knowledge

Spelling

In the following sentences, circle the correctly spelled words within the parentheses.

- Each of the employees attended the meeting (accept except) the line supervisor, who was out of town for job-related travel.
- The (advise advice) he gave will help us all do a better job.
- Management must (affect effect) a change in employees' attitudes toward absenteeism.
- Let me (site cite sight) this most recent case as an example.
- (Its It's) too early to tell if our personnel changes will help create a better office environment.
- If we (lose loose) another good employee to our competitor, our production capabilities will suffer.
- I'm not (quite quiet) sure what she meant by that comment.
- (Their There They're) budget has gotten too large to ensure a successful profit margin.
- We had wanted to attend the conference (to too two), but our tight schedule prevented us from doing so.
- (You're Your) best chance for landing this contract is to manufacture a better product.